

MERN Full Stack Final Assessment (FA)

SET C — Inventory & Procurement Management System

API Response & Frontend Evaluator Contract

1. Purpose of This Document

This document defines the:

- expected API response formats
- status codes
- pagination structures
- authentication response structures
- analytics response formats
- frontend evaluator conventions
- Playwright compatibility requirements

for the:

Inventory & Procurement Management System — Set C

Students must strictly follow:

- route names
- response structures
- status codes
- data-testid conventions
- window.appState structure

mentioned in this document.

Failure to follow these conventions may affect automated evaluation.

2. Common Backend Response Standards

Success Response

Status Code

200 / 201

Expected Structure

```
{
  "success": true,
  "message": "Operation successful",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
    "updatedAt": "ISO_DATE"
  }
}
```

Error Response

Status Codes

Scenario	Status
Validation Error	400
Unauthorized	401
Forbidden	403
Not Found	404
Server Error	500

Expected Structure

```
{
  "success": false,
  "message": "Error message"
}
```

Pagination Response Format

Expected Structure

```
{
  "success": true,
  "message": "Data fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

3. MongoDB & Authentication Routes

POST /sync

Status Code

200

Auth Required

Yes

Allowed Roles

admin

procurement_manager

Expected Response

```
{
  "success": true,
  "message": "Dataset synchronized successfully",
  "data": {
    "totalFetched": 120,
    "inserted": 95,
    "duplicates": 8,
    "rejected": 17
  }
}
```

Values are dynamic based on dataset.

GET /health**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Database connected successfully",
  "data": {
    "database": "connected",
    "documentCount": 95
  }
}
```

POST /auth/register**Status Code**

201

Expected Response

```
{
  "success": true,
  "message": "User registered successfully",
  "data": {
    "_id": "string",
    "userId": "USR101",
    "name": "Dinesh",
  }
}
```

```
{
  "email": "dinesh@test.com",
  "role": "warehouse_staff",
  "createdAt": "ISO_DATE"
}
```

POST /auth/login

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Login successful",
  "token": "JWT_TOKEN",
  "data": {
    "_id": "string",
    "userId": "USR101",
    "name": "Dinesh",
    "email": "dinesh@test.com",
    "role": "procurement_manager"
  }
}
```

GET /auth/me

Status Code

200

Auth Required

Yes

Expected Response

```
{
  "success": true,
  "message": "Authenticated user fetched successfully",
  "data": {
    "_id": "string",
    "userId": "USR101",
    "name": "Dinesh",
    "email": "dinesh@test.com",
    "role": "warehouse_staff"
  }
}
```

4. User APIs

GET /users

Status Code

200

Pagination Supported

Yes

Expected Response

```
{
  "success": true,
  "message": "Users fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string",
      "userId": "USR101",
      "name": "Dinesh",
      "email": "dinesh@test.com",
      "role": "warehouse_staff",
      "status": "active"
    }
  ]
}
```

GET /users/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "User fetched successfully",
  "data": {
    "_id": "string",
    "userId": "USR101",
    "name": "Dinesh",
    "email": "dinesh@test.com",
    "role": "warehouse_staff",
    "department": "Inventory",
    "status": "active"
  }
}
```

```
}  
}
```

5. Supplier APIs

POST /suppliers

Status Code

201

Expected Response

```
{  
  "success": true,  
  "message": "Supplier created successfully",  
  "data": {  
    "_id": "string",  
    "supplierId": "SUP101",  
    "name": "ABC Electronics",  
    "category": "Electronics",  
    "status": "active",  
    "createdAt": "ISO_DATE"  
  }  
}
```

GET /suppliers

Status Code

200

Expected Response

```
{  
  "success": true,  
  "message": "Suppliers fetched successfully",  
  "data": [  
    {  
      "_id": "string",  
      "supplierId": "SUP101",  
      "name": "ABC Electronics",  
      "status": "active"  
    }  
  ]  
}
```

GET /suppliers/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Supplier fetched successfully",
  "data": {
    "_id": "string",
    "supplierId": "SUP101",
    "name": "ABC Electronics",
    "email": "abc@test.com",
    "phone": "9876543210",
    "category": "Electronics",
    "status": "active"
  }
}
```

PATCH /suppliers/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Supplier updated successfully",
  "data": {
    "_id": "string",
    "supplierId": "SUP101",
    "status": "active",
    "updatedAt": "ISO_DATE"
  }
}
```

DELETE /suppliers/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Supplier deleted successfully"
}
```

```
}
```

6. Product APIs

POST /products

Status Code

201

Expected Response

```
{
  "success": true,
  "message": "Product created successfully",
  "data": {
    "_id": "string",
    "productId": "PROD101",
    "name": "Dell Monitor",
    "category": "Electronics",
    "stockQuantity": 20,
    "status": "active",
    "createdAt": "ISO_DATE"
  }
}
```

GET /products

Status Code

200

Pagination Supported

Yes

Expected Response

```
{
  "success": true,
  "message": "Products fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string",
      "productId": "PROD101",
      "name": "Dell Monitor",
      "stockQuantity": 20,

```

```
    "status": "active"
  }
]
}
```

GET /products/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Product fetched successfully",
  "data": {
    "_id": "string",
    "productId": "PROD101",
    "name": "Dell Monitor",
    "category": "Electronics",
    "stockQuantity": 20,
    "reorderLevel": 5,
    "unitPrice": 12000,
    "supplier": {},
    "status": "active"
  }
}
```

PATCH /products/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Product updated successfully",
  "data": {
    "_id": "string",
    "productId": "PROD101",
    "stockQuantity": 25,
    "updatedAt": "ISO_DATE"
  }
}
```

DELETE /products/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Product deleted successfully"
}
```

7. Purchase Order APIs

POST /orders**Status Code**

201

Expected Response

```
{
  "success": true,
  "message": "Purchase order created successfully",
  "data": {
    "_id": "string",
    "orderId": "ORD101",
    "supplier": {},
    "totalAmount": 50000,
    "status": "pending",
    "createdAt": "ISO_DATE"
  }
}
```

GET /orders**Status Code**

200

Pagination Supported

Yes

Expected Response

```
{
  "success": true,
  "message": "Orders fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
}
```

```
"totalPages": 5,  
"data": [  
  {  
    "_id": "string",  
    "orderId": "ORD101",  
    "status": "pending",  
    "totalAmount": 50000  
  }  
]  
}
```

GET /orders/:id

Status Code

200

Expected Response

```
{  
  "success": true,  
  "message": "Order fetched successfully",  
  "data": {  
    "_id": "string",  
    "orderId": "ORD101",  
    "supplier": {},  
    "products": [],  
    "totalAmount": 50000,  
    "status": "approved"  
  }  
}
```

PATCH /orders/:id

Status Code

200

Expected Response

```
{  
  "success": true,  
  "message": "Order updated successfully",  
  "data": {  
    "_id": "string",  
    "orderId": "ORD101",  
    "status": "approved",  
    "updatedAt": "ISO_DATE"  
  }  
}
```

```
}  
}
```

DELETE /orders/:id**Status Code**

200

Expected Response

```
{  
  "success": true,  
  "message": "Order deleted successfully"  
}
```

8. Procurement Workflow APIs

PATCH /orders/:id/approve**Status Code**

200

Expected Response

```
{  
  "success": true,  
  "message": "Purchase order approved successfully",  
  "data": {  
    "orderId": "ORD101",  
    "status": "approved",  
    "approvedBy": {  
      "_id": "string",  
      "name": "Manager"  
    }  
  }  
}
```

PATCH /orders/:id/status**Allowed Status**

pending

approved

shipped

delivered

cancelled

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Delivery status updated successfully",
  "data": {
    "orderId": "ORD101",
    "status": "delivered",
    "updatedAt": "ISO_DATE"
  }
}
```

Workflow Validation Rules

- cancelled orders cannot move to delivered
- delivered orders automatically update inventory
- suppliers cannot approve orders
- warehouse staff cannot cancel approved orders

9. Filtering, Search & Pagination APIs

GET /products?category=Electronics**GET /products?stockMin=10****GET /products?status=active****Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Products filtered successfully",
  "data": [
    {
      "_id": "string",
      "productId": "PROD101",
      "category": "Electronics"
    }
  ]
}
```

GET /suppliers?status=active

GET /suppliers?category=Hardware

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Suppliers filtered successfully",
  "data": [
    {
      "_id": "string",
      "supplierId": "SUP101",
      "status": "active"
    }
  ]
}
```

GET /orders?page=1&limit=10

GET /products?search=monitor

GET /products?category=Electronics&status=active&page=2&limit=5

Status Code

200

Expected Pagination Response

```
{
  "success": true,
  "message": "Products fetched successfully",
  "page": 2,
  "limit": 5,
  "total": 50,
  "totalPages": 10,
  "data": [
    {
      "_id": "string",
      "productId": "PROD101",
      "name": "Dell Monitor"
    }
  ]
}
```

10. Analytics APIs

GET /analytics/inventory

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Inventory analytics fetched successfully",
  "data": {
    "totalProducts": 150,
    "lowStockProducts": 12,
    "outOfStockProducts": 3,
    "activeSuppliers": 18
  }
}
```

Values are dynamic.

GET /analytics/procurement

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Procurement analytics fetched successfully",
  "data": {
    "totalOrders": 90,
    "deliveredOrders": 70,
    "cancelledOrders": 5,
    "monthlyProcurementCost": 450000
  }
}
```

Values are dynamic.

GET /analytics/suppliers

Status Code

200

Expected Response

```
{
  "success": true,
```

```

"message": "Supplier analytics fetched successfully",
"data": [
  {
    "supplier": "ABC Electronics",
    "orderCount": 22,
    "highestProcurementValue": 150000,
    "delayedDeliveries": 2,
    "activeSupplierPercentage": 82
  }
]
}

```

Values are dynamic.

11. Frontend Evaluator Contract

Mandatory React Router Routes

Route	Purpose
/login	Authentication
/dashboard	Analytics dashboard
/suppliers	Supplier management
/products	Product management
/orders	Purchase order management
/inventory	Inventory tracking
/profile	Current user profile

Mandatory window.appState Structure

Students must expose:

window.appState

Expected Structure

```

window.appState = {
  authUser,
  token,
  suppliers,
  products,
  orders,
  inventoryTransactions,
  filters,
  analytics
}

```